The University of New South Wales

School of Electrical Engineering and Telecommunications

ELEC3004: Signal Processing and Transform Methods

Final Examination, Session 1, 2006

Candidates should attempt ALL questions

1. Time allowed: **2** hours.

2. The paper contains **3** questions.

3. The questions are NOT of equal value. Part marks are as indicated .

4. The examination paper may be retained by the candidate.

5. Casio $f\chi$ - 911W scientific calculator provided by examination section.

6. Transform tables are supplied for your use.

ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY ONLY BE USED FOR DRAWING, SKETCHING OR GRAPHICAL WORK.

Question 1. (45 marks)

A. (i) Determine the transfer function $H(z)$ of a discrete-time system from the pole-zero map shown in Figure 1. **[2 marks]**

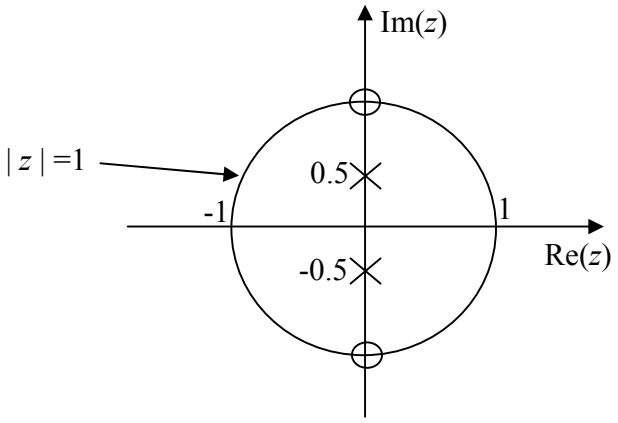


Figure 1

(ii) Sketch the approximate magnitude response for the pole-zero map given in Figure 1 **[3 marks]**

B. Determine the coefficients of a linear-phase FIR digital filter given by equation (1) such that its frequency response is normalised so that H(0) = 1.

$$y[n] = a_0 x[n] + (1/3)\, x[n-1] + a_2\, x[n-2] \qquad (1)$$

[6 marks]

C. By determining the transfer functions for Figure 2, show that the systems (i) and (ii) are equivalent. **[10 marks]**

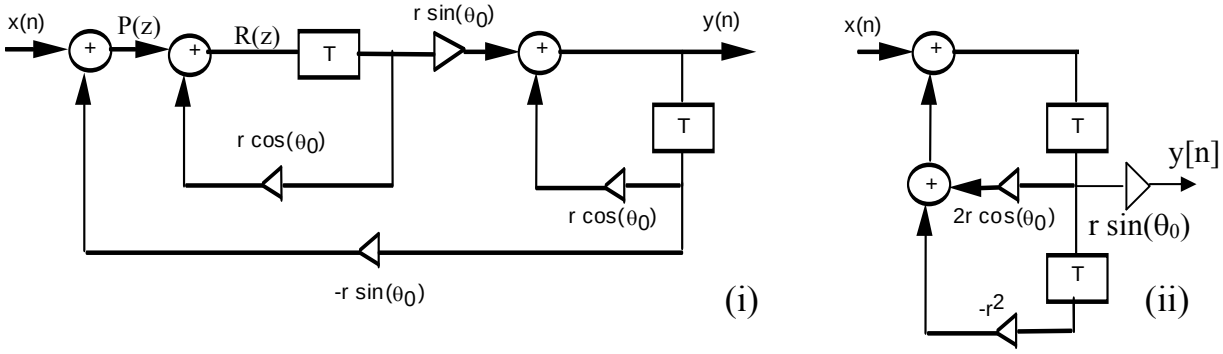


Figure 2

D. (i) A FIR filter has an impulse response as shown in Figure 3.

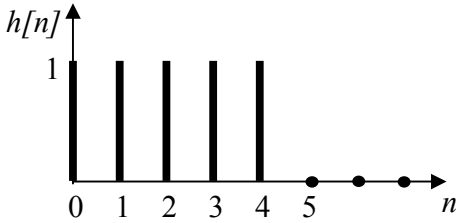


Figure 3

Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{\theta}{2}\right)}$$ where θ is the digital frequency.

[5 marks]

(ii) Draw an implementation of this filter using pole and zero configuration.

[4 marks]

(iii) Show that the above filter has a constant group delay. **[3 marks]**

E. Determine and sketch the magnitude and phase responses of the system shown in Figure 4.

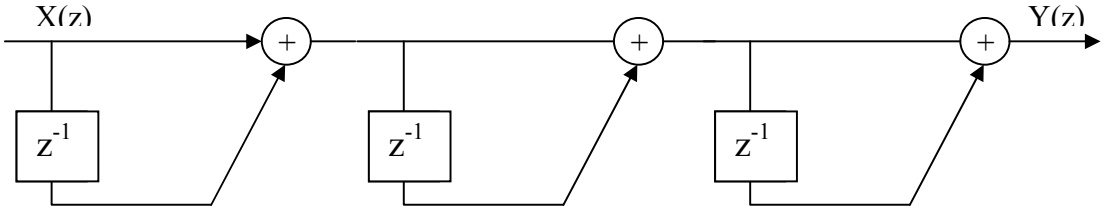


Figure 4

[6 marks]

F. A two-pole digital lowpass filter is given by

$$H(z) = \frac{k^2{}_0}{(1 - b_1 z^{-1})^2}$$

Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad \left|H(\theta)\right|^2_{\theta=\frac{\pi}{3}} = \frac{1}{4}$$

[6 marks]

Question 2. (30 marks)

A. A digital oscillator that simultaneously generates cosine and sine waves is required for telecommunications applications. Using the following equations you are required to design an oscillator with an oscillation frequency of θ .

$$\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos(n\theta)$$

$$\cos(n+1)\theta - \cos(n-1)\theta = -2\sin\theta\sin(n\theta)$$

(i) Determine the difference equations for the digital oscillator. **[6 Marks]**

(ii) Draw a structure for this oscillator with 4 multipliers, where two of the multipliers are –1. **[6 Marks]**

(iii) Determine the coefficients required to obtain the desired oscillation frequency of $3\pi/4$. **[2 Marks]**

B. The sampling period T of the input signal shown in Figure 5 is 125 μs. The relative frequency is θ = ωT. The first oscillator generates a carrier with a relative frequency $\theta_1 = \omega_1 T$, where $\omega_1 = 2\pi.2.10^3\ rad/\sec$. The second oscillator generates a carrier with a relative frequency $\theta_2 = (\omega_c + \omega_1)\,T$, where $\omega_c = 2\pi.10^3\ rad/\sec$. The low-pass filter has the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \dfrac{\pi}{4} \\ 0 & \dfrac{\pi}{4} < |\theta| \le \pi \end{cases}$$

If the input spectrum is $X(\theta) = 1 + \cos(\theta) \quad -\pi \le \theta \le \pi$,

(i) Write appropriate equations for $B_c(\theta)$, $D_c(\theta)$, and $R_c(\theta)$. **[8 Marks]**

(ii) Sketch |X(θ)|, $|B_c(\theta)|$, $|D_c(\theta)|$, and $|R_c(\theta)|$ against θ, **[8 Marks]**

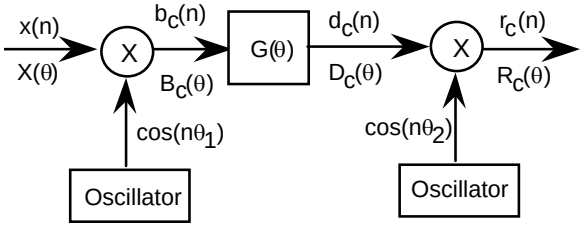


Figure 5

Note: $x(n)\cos(\theta_0 n) \quad \Leftrightarrow \quad \frac{1}{2}\left[X(\theta-\theta_0) + X(\theta+\theta_0)\right]$

Question 3. (25 marks)

A. It is required to design a digital filter to approximate the analogue filter given by the following transfer function

$$H(s) = \frac{0.1s}{s^2 + 0.1s + 1}$$

(i) Using the bilinear transform show that the resulting digital filter can be written in the form of

$$H(z) = \frac{(a_0 - a_0 z^{-2})}{1 + b_2 z^{-2}}$$

Determine the values of the coefficients a_0 and b_2. The resulting digital filter is to have a resonant frequency of 600Hz. The sampling frequency is 2400 Hz. **[7 marks]**

(ii) Draw a canonic realisation of $H(z)$. **[4 marks]**

B. Let a causal linear time invariant discrete-time system be characterised by a real impulse response *h[n]* with a DTFT of $H(\theta)$. Consider the system of Figure 6, where *x[n]* is a finite length sequence. Determine the frequency response of the overall system G(θ) in terms of $H(\theta)$, and show that it has a zero-phase response. **[7 marks]**

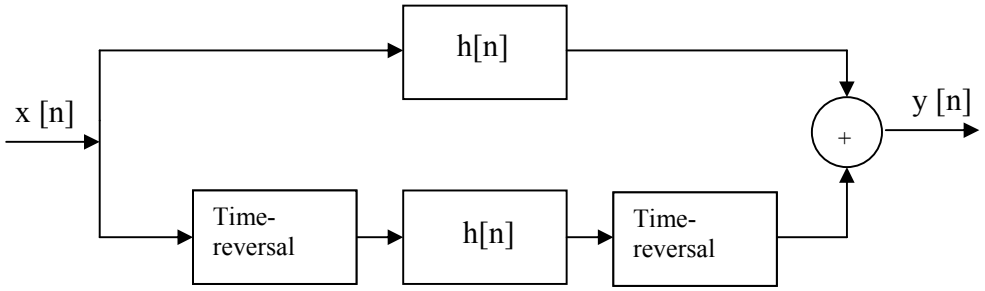


Figure 6

C. A signal has been sampled at 44 kHz and for a particular application this signal is required to be used at a reduced sampling rate of 8 kHz.

(i) Draw a block diagram to show how you would achieve this. **[4 marks]**

(ii) Indicate the type of digital filter which you would choose for part (i) above (give reasons). **[3 marks]**

Table of the z transforms of useful sequences

Basic z-Transforms	**Sequence**	**z-Transform**
1. Unit sample	$\delta[n]$	1
	$\delta[n-k]$	z^{-k}
2. Unit step	$u[n]$	$z/(z-1)$
3. Exponential	$a^n u[n]$	$z/(z-a)$
4. Sinusoidal	$\sin[\theta_o n]\, u[n]$	$\dfrac{z \sin\theta_0}{z^2 - 2\cos\theta_0\, z + 1}$
	$\cos[\theta_o n]\, u[n]$	$\dfrac{z^2 - z\cos\theta_0}{z^2 - 2\cos\theta_0\, z + 1}$
5. Unit ramp	$nu[n]$	$\dfrac{z}{(z-1)^2}$

$$1 + z^{-1} + z^{-2} + z^{-3} + \ldots\ldots\ldots + z^{-(N-1)} = \frac{1 - z^{-N}}{1 - z^{-1}}$$

z-Transform Properties	**Sequence**	**z-Transform**
	$x[n]$	$X(z)$
	$x[n-k]$	$z^{-k} X(z)$
	$a^n x[n]$	$X(z/a)$
	$x[-n]$	$X(z^{-1})$
	$x[n]*y[n]$	$X(z).Y(z)$
	$n\, x[n]$	$-z\, (dx(z)/dz)$

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\cos\theta = \frac{1}{2}\left[e^{j\theta} + e^{-j\theta}\right]$$
$$\sin\theta = \frac{1}{2j}\left[e^{j\theta} - e^{-j\theta}\right]$$

Fourier Series:

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t); \qquad \omega_0 = 2\pi f_0; \quad f_0 = \frac{1}{T}$$

The fundamental frequency of x(t) is ω_0 and T is the period of the signal x(t).

$$A_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\cos(n\omega_0 t)\,dt \qquad n = 0,1,2.... \qquad B_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\sin(n\omega_0 t)\,dt \qquad n = 1,2,3...$$

Fourier Transform pair (for discrete-time signal)

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}... \qquad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

$\theta = digital\ frequency; \quad -\pi \le \theta \le \pi$

A **Butterworth filter** of order'n' is defined by

$$|H(j\omega)|^2 = \frac{1}{1+\left(\frac{\omega}{\omega_C}\right)^{2n}}$$ where ω_c is the cut-off frequency of the filter

n	Butterworth Polynomials
1	$S + 1$
2	$s^2 + \sqrt{2}s + 1$
3	$(s^2 + s + 1)(s + 1)$

Impulse Invariant Method:

$$\frac{1}{s+b} \rightarrow \frac{1}{1 - e^{-bT} z^{-1}} \qquad b > 0$$ *T*-sampling period

Bilinear Transformation:

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}} \qquad T - sampling \quad period$$

End of paper